MIL-C-49206(CR)

7 May 1980

MILITARY SPECIFICATION

CONTROL, COMMUNICATIONS MODE SELECTOR, CMSC, C-10377()/GTC

This specification is approved for use by US Army Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Communications Mode Selector Control, CMSC, C-10377()/GTC which is a tactical, ruggedized applique device which interfaces the TSEC/KY-57 with remote wire-line users, local data users and local handsets. It provides for automatic mode switching of the TSEC/KY-57 and for proper routing of received data to the appropriate user interface using a half duplex link configuration. (See 6.7)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall be considered a superseding requirement.

SPECIFICATIONS

NN-P-71	Pallets, Material Handling, Wood, Stringer Construction, 2-way and 4-way (Partial)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to US Army Communications Research and Development Command, ATTN: DRDCO-CM-DM, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5810

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PPP-B-585	Boxes, Wood Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated Triple-Wall
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	Strapping, NonMetallic (and Connectors)
PPP-T-97	Tape, Packaging, Industrial Filament Reinforced
QQ-S-781	Strapping, Steel, and Seals

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-S-901	Shock Tests, H.I. (HI Impact) Shipboard Machinery, Equipment and Systems, Requirement for
MIL-P-11268	Parts, Materials and Processes used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Flexible, for Packaging

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STANDARDS

Federal

FED-STD-356	Commercial Packaging of Supplies and Equipment

Military

MIL-STD-105	Standard Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Load for 40" x 48" Pallets
MIL-STD-188-114	Digital Interface Circuits, Electrical Characteristics of
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461 Notice 4	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-721	Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety
MIL-STD-781B	Reliability Tests; Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipments and Facilities
MIL-STD-1562	Lists of Standard Microcircuits

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DRAWINGS

National Security Agency

ON190303	KYX-15/TSEC
ON217339	TSEC/KY-57, Back to Back Tester with Cables
ON241700	TSEC/KY-57
ON241780	Vehicular Power Adapter HYP-57/TSEC
ON241845	Z-AIJ Battery Case

CORADCOM

DLB4000090	CMSC Assembly
B4000184	Interface Control Drawing

PUBLICATIONS

*National Security Agency

CSESD-14	Communications Security Equipment System Document for TSEC/KY-57/58 (U)

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and identifying number or symbol should be stipulated when requesting copies.)

*NACSEM 5100 Compromising Emanations Laborator Test Standard, Electromagnetics (U). (Application for copies should be addressed to the procuring activity or as directed by the contracting officer.)

*Classified Document

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

ANSI z35.1 Accident Prevention Signs, Specification For

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY, 10018.)

Title 29, Chapter XVII, Part 1910, code of Federal Regulations, Occupational Safety and Health

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

National Institute for Occupational Safety and Health (NIOS-H), Toxic Substance Annual List

(Application for copies should be addressed to US Department of Health, Education and Welfare, Rockville, Maryland 20852.)

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3. REQUIREMENTS

3.1 First Article. When specified by the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed to conform with the requirements of this specification, and of Drawing DL-B4000090 including the requirements for parts, materials, and processes thereon.

3.3 Parts, materials, and processes; general. (See 4.4). In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Preconditioning. (See 4.7.)

3.4.1 Bounce preconditioning. (See 4.7.1) Each assembled CMSC shall meet the requirements specified herein without subsequent processing, after subjection to the bounce preconditioning of 4.7.1.

3.4.2 Electrical preconditioning. (Burn-in) (See 4.7.2) Each assembled CMSC shall be subjected to electrical preconditioning (burn-in) for a period of 48 hours (equipment "on" time) of which the last 24 hours shall be failure free.

3.5 C-10377 Performance Requirements. The CMSC shall provide the following basic modes of operation.

3.5.1 Transmission of local digital data. (See 4.8.4.1.) (Refer to 6.4 and 6.11 for terminology and interface definitions.) The CMSC shall recognize keying signals coming from a data device. The CMSC front panel mode switch when in the digital data position and/or with the local DDMC grounded, will identify a digital data transmission. The CMSC shall provide keying and DDMC (Digital Data Mode Control) signals to the TSEC/KY-57. The TSEC/KY-57 as defined by CSESD-14 will send out its normal synchronization preamble. As soon as the TSEC/KY-57 is ready to accept digital data, the DDCO (Digital Data Clock Out) line will become active. The CMSC shall detect this and at this time insert a data flag into the data stream indicating that the following information is data and should be processed accordingly. The DDCO signal coming from the TSEC/KY-57 shall now be applied to the data device and data clocked in. At the termination of the data transmission the data device keying signal will go inactive. The CMSC shall recognize this and cause the TSEC/KY-57 keying line to go inactive. The TSEC/KY-57 will terminate the transmission with an EOM (End of Message).

3.5.1.1 Data Flag. (See 4.8.4.1.) The data flag shall be an Octal 7-5-3-1-0 code if the message to be transmitted is digital data, 1200/2400 Hz FSK or analog 1 data. The flag shall consist of 30 repeats of the above code followed by one inverted code. Satisfactory completion of the test of 4.8.4.1 verifies that the proper data flag was transmitted.

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3.5.1.1.1 Flag Detection. (See 4.8.4.1) A receive data flag shall be valid if at least one perfect frame (15 bits - either inverted or not) with three successive frames, having five or less errors, is received.

3.5.1.2 Digital data and clock levels. (See 4.8.4.1 and 4.18) The output digital data and the output clock levels shall be +6V ±1V, -6V ±1V when driving a load impedance of 6000 ohms minimum and as defined in MIL-STD-188-114.

3.5.1.3 Local Voice Interrupt. (See 4.8.4.1.) During the course of a data transmission the operator shall have the capability to interrupt the data transmission and transmit a voice message. This shall be accomplished by the operator initiating a double PTT on the handset. The CMSC shall recognize the double PTT coming from the handset and cause the PTT going to the TSEC/KY-57 to go inactive. Sufficient time will be permitted for the KY-57 to transmit an EOM. The CMSC shallnow initiate a new transmission in the local voice mode.

3.5.1.4 Power Consumption. (See 4.8.4.1.) Average power consumption shall not exceed 3 watts from a nominal +24 VDC source for a 50% transmit/50% receive duty cycle.

3.5.2 Reception of Digital Data. (See 4.8.4.2.) The TSEC/KY-57 will recognize incoming 16 kilobit information and power up. The CMSC shall recognize the power-up condition and place the TSEC/KY-57 in the digital data mode. The TSEC/KY-57 will synchronize in the normal fashion. At the end of the synchronization preamble, the receive digital clock will be available at DDCO on the TSEC/KY-57 and the data flag shall be recognized by the CMSC with data routed to the local data connector since the digital mode and/or the local DDMC are set. The TSEC/KY-57 shall be held in the digital data mode until it powers down. The power-down state can be recognized at the audio connector.

3.5.2.1 Data Inversion Correction. (See 4.8.4.2.) The receive CMSC shall use the transmitted data flag to determine and correct data inversions caused by the transmission path.

3.5.3 Transmission of Local Analog 1 Data. (See 4.8.4.3.) Analog 1 data is that data processed thru the A/D converter of the TSEC/KY-57. The CMSC shall recognize a keying signal coming from a data device. The CMSC front panel mode switch, when in the Analog 1 Data position, and/or the local ADMC is grounded, will identify an Analog 1 Data transmission. The CMSC shall provide the keying and DDMC signals to the TSEC/KY-57. The TSEC/KY-57 will send out its normal synchronization preamble. As soon as the TSEC/KY-57 is ready to accept the data flag, the DDCO line will become active. The CMSC shall, at this time, insert a data flag indicating that the following information is data and should be processed accordingly. The CMSC shall now place the transmit TSEC/KY-57 in the Analog Data mode. At the termination of the data transmission, the data device keying signal will go inactive. The CMSC shall recognize this and cause the TSEC/KY-57 PTT (push-to-talk) line to go inactive. The TSEC/KY-57 will terminate the transmission with an EOM (End of Message) and power down.

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3.5.3.1 Outpower Level. (See 4.8.4.3.) The A1 output level to the TSEC/KY-57 as measured at the DMIC terminal shall be within the range of 0.1 to 1000 mV.

3.5.4 Reception of Analog 1 Data. (See 4.8.4.4.) The TSEC/KY-57 will recognize 16 Kbit information and power up. The CMSC shall recognize the power-up condition and place the TSEC/KY-57 in the digital data mode. The TSEC/KY-57 will synchronize in the normal fashion. At the end of the synchronization preamble, the receive digital data clock will be available at the DDCO on the TSEC/KY-57 and the data flag shall be recognized. The front panel mode switch must be in the Analog 1 Data position and/or the local ADMC grounded. The CMSC shall take the TSEC/KY-57 out of the digital data mode and place it in the analog data mode and route data to the local data connector until the TSEC/KY-57 powers down.

3.5.4.1 Outpower Level. (See 4.8.4.4.) The A1 output level to the local data user shall be 3V RMS, +3, -6 dB.

3.5.5 Transmission of Local or Remote (Wireline) Analog 2 Data. (See 4.8.4.5.) Analog 2 Data is defined as that analog data which is more faithfully reproduced by not putting it through the A/D converter of the TSEC/KY-57 but by amplifying it and processing it as digital data for transmission. This processing consists of converting the low level analog data signal sufficiently to allow it to be processed as digital data by the TSEC/KY-57. For local A2 data transmission, the CMSC shall recognize a keying signal coming from the local data device and the CMSC front panel mode switch shall be in the Analog Data 2 position. For transmission of A2 data from the wireline user, the CMSC front panel mode switch shall be in the 3900 or VOX position and the CMSC shall derive a PTT from a 3900 Hz tone or A2 data on the wireline via an internal VOX. The TSEC/KY-57 will send out its normal synchronization preamble. As soon as the TSEC/KY-57 is ready to accept the data flag, the DDCO line will become active. The CMSC shall, at this time, insert a data flag indicating that the following information is data and should be processed accordingly. The CMSC shall now route the analog 2 signal to the TSEC/KY-57 digital data port and process it as digital data. At the termination of the data transmission the keying signal will go inactive and the TSEC/KY-57 will terminate the transmission with an EOM.

3.5.6 Reception of Analog 2 Data. (See 4.8.4.6.) The TSEC/KY-57 will recognize its incoming 16 Kbit information and power up. The CMSC shall recognize the power-up condition and place the CMSC in the digital data mode. The TSEC/KY-57 will synchronize in the normal fashion. At the end of the synchronization preamble, the receive digital clock will be available at DDCO on the TSEC/KY-57 and the data flag shall be recognized. The front panel switch being in the Analog Data 2 position shall be recognized and the received digital message shall be filtered and output to the local data device. To route received data to a wireline user, the front panel mode switch must be in VOX or 3900.

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3.5.6.1 Output Level. (See 4.8.4.6.) The A2 output to the local data connector shall be 0.21V ±0.07V.

3.5.7 Transmission of Voice. (See 4.8.4.7.)

3.5.7.1 Transmission of Wireline Voice. (See 4.8.4.7.1, 4.8.4.7.2.) The CMSC shall derive a PTT from a 3900 Hz tone or via a VOX circuit, provide a PTT to the TSEC/KY-57 and route the voice to the proper TSEC/KY-57 terminations. The TSEC/KY-57 will then process the voice message as it normally would do without a CMSC.

3.5.7.2 Transmission of Local Voice. (See 4.8.4.7.3.) The CMSC shall be provided a local PTT via the local data connector with the mode switch in the "A2" position. The CMSC A2 data/voice detector shall recognize voice and transmit a PTT and voice to the TSEC/KY-57 for normal voice processing.

3.5.7.3 Transmission of Handset Voice. (See 4.8.4.7.4) The CMSC shall recognize a PTT signal coming from the attached handset. The CMSC shall provide a PTT to the TSEC/KY-57 and connect the handset to the proper TSEC/KY-57 terminations. The TSEC/KY-57 will process the voice message as it normally would do without a CMSC.

3.5.8 Reception of Voice. (See 4.8.4.8.) The TSEC/KY-57 will recognize 16 Kbit cipher information and power up. The CMSC shall recognize the power-up condition and place the TSEC/KY-57 in the digital data mode. The TSEC/KY-57 will synchronize in the normal fashion. At the end of the synchronization preamble the CMSC shall search for the data flag. Upon not finding a data flag, the CMSC shall place the TSEC/KY-57 into the normal voice mode for the remainder of the message and direct voice to the appropriate point identified by the front panel mode switch. If a system power failure occurs, all transmissions and receptions will be terminated and communications will revert back to a local handset mode. If because of channel errors the data flag is not detected, the CMSC will always default to the voice mode.

3.5.8.1 CMSC Frequency Response. (See 4.8.4.8.) The received CMSC voice output level when measured over the frequency range of 300 Hz to 2700 Hz shall be 0 dBm ±4 dBm.

3.5.9 Plain Text Reception/Routing. (See 4.8.4.9.) The CMSC, with DC power on, when not transmitting or receiving shall monitor the TSEC/KY-57 AO/DDO line for plain text activity. When activity is detected and no other TSEC/KY-57 control signals are present, the CMSC shall route the plain text to the remote wireline or to the local data connector depending on the remote or local switch setting of the front panel Mode Control switch.

3.5.10 Power off, Bypass Mode. (See 4.8.4.10.) With the DC power switch off, the handset shall have direct electrical connections to the TSEC/KY-57 audio connector.

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3.5.11 Priority Mode Control. (See 4.8.4.1.2, 4.8.4.2.2, 4.8.4.3.2, 4.8.4.4.2) Control signals (ground) on either the local data connector's Analog Data Mode Control (LADMC) or Digital Data Mode Control (LDDMC) causes the mode function switch to be overridden and the CMSC placed in the mode indicated by the grounded pin. However, the Push-to-Talk (PTT) must be released or the reception completed before the system can be properly set up according to the control signal.

3.6 Service conditions. (See 4.9) The equipment shall meet the following service conditions.

3.6.1 Temperature. (See 4.9.1.)

3.6.1.1 Low. (See 4.9.1.1.) During and after testing as specified in 4.9.1.1, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive.

3.6.1.2 High. (See 4.9.1.2.) During and after testing as specified in 4.9.1.2, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive.

3.6.2 Humidity. (See 4.9.2.) During and after testing as specified in 4.9.2, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive. At the conclusion of this test, the equipment shall exhibit no physical damage such as corrosion, rust, blistering, swelling or deterioration of parts and materials.

3.6.3 Low Pressure (Altitude). (See 4.9.3.) During testing as specified in 4.9.3, the equipment shall meet the requirements specified in 3.5.1 and verified per 4.8.4.1.1. After testing, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive.

3.6.4 Vibration, Shock and Bounce. (See 4.9.4)

3.6.4.1 Vibration. (See 4.9.4.1) After testing as specified in 4.9.4.1, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive.

3.6.4.2 Shock Hi Impact. (See 4.9.4.2) During testing as specified in 4.9.4.2 the equipment shall meet the requirements specified in 3.5.1 and verified per 4.8.4.1.1. After testing, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive

3.6.4.3 Shock (Bench Handling). (See 4.9.4.3) After testing as specified in 4.9.4.3, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive.

3.6.4.4 Vibration Secure Cargo. (See 4.9.4.4) After testing as specified in 4.9.4.4, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive.

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3.6.4.5 Bounce, Vehicular. (See 4.9.4.5) After testing as specified in 4.9.4.5, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive.

3.6.4.6 Bounce, Loose Cargo. (See 4.9.4.6) After testing as specified in 4.9.4.6, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive.

3.6.5 Salt Fog. (See 4.9.5) After testing as specified in 4.9.5, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive, and when examined visually with the aid of a 10 power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering, or loosening of finish or metal surfaces or exfoliation (See 6.4.2) of metal.

3.6.6 Fungus. (See 4.9.6) After the fungus test specified in 4.9.6, the equipment shall be visually examined, using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growith (see 6.4.4) as evidenced by growth colonization (see 6.4.3) (which includes branching (see 6.4.1) and sporulation (see 6.4.5) on or within each cubic foot or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.4.12) shall not be included in this evaluation.

3.6.7 Temperature shock. (See 4.9.7) After testing as specified in 4.9.7, the unit shall meet the requirements of 3.5.1 through 3.5.10 inclusive.

3.6.8 Sunshine. (See 4.9.8) After testing as specified in 4.9.8, the equipment shall meet the requirements of 3.5.1 through 3.5.10 inclusive.

3.6.9 Rain. (See 4.9.9) After testing as specified in 4.9.9, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive and shall, upon internal inspection, contain no water.

3.6.10 Explosive Atmosphere. (See 4.9.10) During testing as specified in 4.9.10, the equipment shall not cause explosion.

3.6.11 Immersion. (See 4.9.11) After testing as specified in 4.9.11, the equipment shall not contain any water.

3.6.12 Dust. (See 4.9.12) After testing as specified in 4.9.12, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive.

3.6.13 Shock Drop. (See 4.9.13) After testing as specified in 4.9.13, the equipment shall meet the requirements specified in 3.5.1 through 3.5.10 inclusive.

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3.7 Electromagnetic Interference (EMI) - First Article. (See 4.10) The CMSC shall comply with the following emission and susceptibility requirements of MIL-STD-461, Notice 4:

CE01	CS01	RE02	RS02
CE03	CS02	RE02.1(Note 1)	RS03(Note 2)
CE04	CS06		RS03.1(Note 2)
CE05			

NOTE 1 - Upper frequency limit of test for RE02.1 shall be 1000 MHz.

NOTE 2 - The equipment shall be immersed in electric fields of the following intensity using the specified modulation characteristics.

FREQ. RANGE	FLD INTENSITY	MODULATION CHARACTERISTICS
.010-1.9 MHz	1 volt/meter	AM, 400 Hz square wave, 50%
2-75.9 MHz	5 volt/meter	FM, +8 kHz deviation, 1000 Hz
76-399 MHz	10 volt/meter	AM, 1000 Hz sine wave 50%
.400-12.4 GHz	10 volt/meter	pulse, 1 usec pulse width, 400 pps

3.7.1 Electromagnetic Interference - Group C Inspection. (See 4.10) The CMSC shall comply with the following emission and susceptibility requirements of MIL-STD-461, Notice 4:

CE04: 50 KHz to 50 MHz
RE02: 2 MHz to 400 MHz
RE02.1: 2 MHz to 400 MHz
RS03: 2 MHz to 400 MHz
RE03.1: 2 MHz to 400 MHz

3.7.2 Bonds and Grounds. (See 4.10.1) Bonding for equipment current path returns and shock hazards shall be so installed that expansion, contraction, or movement incident to normal service use and maintenance will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high resistance coatings from the immediate areas of contact. Upon completion of the bond and verification of its DC resistance, the equipment finish shall be restored to its original condition in accordance with MIL-F-14072. The DC resistance of bonds and grounds shall not exceed 2.5 milliohms.

3.7.3 Cables and Connectors. (See 4.10.1) All shielded cables shall have the shield terminated in an approved military connector with an EMI backshell to provide bonding of the shield. The use of shielded cables shall be kept to a minimum by the use of alternative suppression techniques such as pin filters. All panel mounted connectors and components and the panel mating surfaces shall be free of nonconductive finishes and provide bonding and grounding with mating connectors and equipment ground plane. The DC bonding resistance for all connectors and panel mounted components shall not exceed 2.5 milliohms.

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3.8 Tempest. (See 4.11) The CMSC shall meet the requirements of NACSEM-5100 when tested in accordance with 4.11.

3.9 Interchangeability. (See 4.12) Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454.

3.10 Safety Engineering. (See 4.6)

3.10.1 Personnel Hazards. (See 4.6) Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

3.10.2 Radioactive Materials. Radioactive materials which result in health hazards shall not be used (e.g., luminous dials/markings, electron tubes).

3.10.3 Warning Labels. (See 4.4) Warning labels, when applicable, shall be provided on the equipment in accordance with ANSI Z35.1-1972.

3.10.4 Edge Rounding. (See 4.4) Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm).

3.11. Reliability. (See 4.14) The equipment shall demonstrate a specified mean-time-between-failures (Θ_0) of 6,000 hours when used in a ground mobile and tracked vehicle environment at 50% transmit/50% receive duty cycle at an ambient temperature of 50 degrees C.

3.12 Subassemblies (Electrical Test). (See 4.13) All CMSC subassemblies shall meet the requirements of their respective drawings when tested in accordance with 4.13.

3.13 Size and Mechanical Interface. (See 4.12) The CMSC shall be the same size and form factor as the TSEC/KY-57. The CMSC shall also be designed to utilize TSEC/KY-57 mounts, vehicular power adaptors and batteries interchangeably. The interfaces shall comply with the requirements of drawings B4000090 and B4000184.

3.14 Weight. (See 4.16) The weight of the CMSC as described on drawing B4000090 shall not exceed 7 pounds.

3.15 TSEC/KY-57 System Test. (See 4.15) The first and each one hundredth CMSC shall be system tested with a TSEC/KY-57 to verify that proper interface controls are being maintained.

3.16 Microelectronic Integrated Circuits. (See 4.4) Microelectronic integrated circuit devices selected for use in equipment redesign shall be selected from MIL-STD-1562. If not included in MIL-STD-1562, they must be submitted for approval as outlined in the contract data requirements list.

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3.17 Finish. (See 4.4) Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, clamps and covers with necessary touchup after mounting. The final paint film on Type I surfaces shall be forest green lusterless enamel in accordance with Table V of MIL-F-14072.

3.18 Marking. (See 4.4)

3.18.1 General. Marking shall conform to MIL-M-13231 unless specified otherwise herein. Front panel marking shall be Group I as described in that specification. Photographically printed aluminum plates shall not be used for front panel marking.

3.18.2 Visibility. Whenever practicable, parts and components shall be so mounted that their identification markings with be readily visible with minimum disassembly of the equipment.

3.18.3 Serial Numbers. The CMSC shall have consecutively assigned serial numbers unless otherwise specified by the contracting officer.

3.19 Workmanship. (See 4.4 and 4.6) The equipment shall be manufactured and assembled in accordance with the requirements of MIL-P-11268 and requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilities for Government Inspections shall be made available by the contractor. The Government reserves the right for authorized technical representatives to witness any contractor performed inspections and tests.

4.2 Classification of Inspection. Inspection shall be classified as follows:

a. First Article Inspection (does not include packaging (see 4.3)).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality Conformance Inspection:

(1) Quality conformance inspection of equipment before packaging (see 4.5).

(2) Quality conformance inspection of packaging (see 4.17).

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4.3 First Article. Unless otherwise specified in the contract or purchase order, the First Article Inspection shall be performed by the contractor.

4.3.1 First Article Units. The contractor shall fabricate 28 first article units of the complete C-10377.

4.3.2 First Article Inspection. The first article inspection shall consist of inspections specified in Table II and shall be performed in the order specified therein except that all hardwired assemblies shall undergo inprocess Group A Inspection (see 4.5.1a) prior to bounce and electrical preconditioning.

4.4 Inspection Covered by Subsidiary Documents. The items in Table I shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

TABLE I - INSPECTION BY SUBSIDIARY DOCUMENTS

ITEM	WHERE REQUIRED
Parts, Materials, and Processes; General	3.3
Finishes	3.17
Marking	3.18, 3.10.3
Microelectronic Integrated Circuits	3.16
Workmanship	3.10.4, 3.19

4.5 Quality Conformance Inspection of Equipment before Packaging. The contractor shall perform the inspections specified in 4.4, 4.7.1, 4.7.2, and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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TABLE II - FIRST ARTICLE INSPECTION

INSPECTION(SEE NOTE A)	REQ PARA	TEST PARA	UNIT (SEE NOTE B) 1	2	3	4	5	6	7	8	9-10	11-26	27-28
1. Inspection covered by subsidiary documents		4.4	Inspection to be performed on all units										
2. Bounce pre-conditioning	3.4.1	4.7.1	Inspection to be performed on all units										
3. Electrical precondi-tioning (burn-in)	3.4.2	4.7.2	Inspection to be performed on all units										
4. Group A inspection	SEE TABLE III		Inspection to be performed on all units										
5. Group B inspection	SEE TABLE IV		Inspection to be performed on all units										
6. Group C inspection													
Low Temperature	3.6.1.1	4.9.1.1	1										
High Temperature	3.6.1.2	4.9.1.2	2										
Humidity	3.6.2	4.9.2		2									
Altitude	3.6.3	4.9.3		1									
Vibration	3.6.4.1	4.9.4.1			1								
Shock Ballistic	3.6.4.2	4.9.4.2			5								
Shock Bench Handling	3.6.4.3	4.9.4.3				1							
Vibration Secure Cargo	3.6.4.4	4.9.4.4			2								
Bounce, Vehicular	3.6.4.5	4.9.4.5			4								
Bounce, loose Cargo	3.6.4.6	4.9.4.6			3								
Salt Fog(See Note C)	3.6.5	4.9.5	3										
Fungus(See Note C)	3.6.6	4.9.6					1						
Temperature Shock	3.6.7	4.9.7						3					
Sunshine	3.6.8	4.9.8						1					
Rain	3.6.9	4.9.9						2					
Explosive At-mosphere	3.6.10	4.9.10						4					
Immersion	3.6.11	4.9.11			6								
Dust(See Note C)	3.6.12	4.9.12				3							
Shock Drop	3.6.13	4.9.13				2							
EMI	3.7	4.10							1				
TEMPEST	3.8	4.11								1			
Safety	3.10	4.6									1		

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TABLE II - FIRST ARTICLE INSPECTION - (CONT'D)

INSPECTION(SEE NOTE A)	REQ PARA	TEST PARA	UNIT (SEE NOTE B) 1	2	3	4	5	6	7	8	9-10	11-26	27-28
7. Group D inspection (Reliability (See Note D)	SEE TABLE VI											1	
8. TSEC/KY-57 System Test	3.15	4.15											1

NOTES:

A. The inspections 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

B. The numbers in the unit columns in the table covered by Inspections 6 through 8 specify the order of inspections for the indicated unit.

C. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

D. Twelve units shall be selected at random for reliability testing, the remaining four units shall be used for contractor training as specified in the contract.

4.5.1 Group A Inspection. Each CMSC on contract or purchase order shall be inspected for conformance to the inspections specified in Table III. The order of inspection of the subassemblies, assemblies and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the CMSC subassemblies shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in Table III.

b. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using general inspection levels and AQL's indicated in Table III (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

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c. Assemblies from the discrete lots which have passed the above Quality Assurance Verification Inspection shall be assembled into the end equipments which shall then be subjected by the contractor to the preconditioning requirements of 4.7 and a 100% visual/mechanical and electrical inspection in accordance with Table III in that order.

NOTE: Assemblies are not required to undergo further visual/mechanical and electrical inspection as part of the contractor's 100% Group A Inspection of the end equipment.

d. After completion of the above 100% inspection by the contractor, discrete lots shall be formed from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQL's indicated in Table III. Group A electrical inspection shall be performed in any order which is satisfactory to the Government, except that the operational test 4.8.4 shall be performed last (see 6.6)

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TABLE III - GROUP A INSPECTION

INSPECTION	RQT PARA	INSP PARA	AQL MAJOR	AQL MINOR
Visual - Mechanical				
Case and Covers	3.19	4.6	1%	4%
PWA Assemblies	3.12	4.13	1%	4%
Electrical Performance				
Transmission of Digital Data	3.5.1	4.8.4.1	1%	*
Reception of Digital Data	3.5.2	4.8.4.2	1%	*
Transmission of A1 Data	3.5.3	4.8.4.3	1%	*
Reception of A1 Data	3.5.4	4.8.4.4	1%	*
Transmission of A2 Data	3.5.5	4.8.4.5	1%	*
Reception of A2 Data	3.5.6	4.8.4.6	1%	*
Transmission of Voice	3.5.7	4.8.4.7	1%	*
Reception of Voice	3.5.8	4.8.4.8	1%	*
Plain Text Reception/Routing	3.5.9	4.8.4.9	1%	*
Power off, Bypass Mode	3.5.10	4.8.4.10	1%	*
Compliance With MIL-STD-188-114**	3.5.1.2	4.18	1%	*

*Defects in all characteristics other than visual-mechanical are considered major.

** This test will be performed only during First Article Tests.

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4.5.2 Group B Inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IV and the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B Sampling Plan The Group B AQL shall be as shown in Table IV and the inspection level shall be S-4.

TABLE IV - GROUP B INSPECTION

INSPECTION	REQT PARA	INSP PARA	AQL
Interchangeability			
Mechanical	3.9	4.12	6.5%
Size and Mechanical Interface	3.13	4.12	6.5%
Weight	3.14	4.16	4.0%
Finish	3.17	4.4	6.5%

4.5.2.2 Order of Inspection Within Group B. Group B inspection shall be performed in any order satisfactory to the Government

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table V. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C Inspection.

4.5.3.1.1 Subgroup 1. For this subgroup, five units shall be selected at random from the first production lot. For subsequent Group C inspection, five units shall be selected from each 600 production units. One unit shall be subjected to Low Temperature, High Temperature, Altitude, and Humidity Testing; one unit shall be subjected to Vibration Secure Cargo, Vibration, Vehicular Bounce, Ballistic Shock, and Immersion Testing; one unit shall be subjected to Bench Handling Shock, Loose Cargo Bounce, and Shock Drop Testing; one unit shall be subjected to EMC Testing; and one unit shall be subjected to TEMPEST Testing.

4.5.3.1.2 Subgroup 2. For this subgroup, three units shall be selected at random from the first production lot. For subsequent Group C inspection, three units shall be selected from each 1000 production units. One unit shall be subjected to Temperature Shock, Sunshine, Rain, and Explosive Atmosphere Testing; one unit shall be subjected to Salt Fog and Dust Testing; and one unit shall be subjected to Fungus Testing.

4.5.3.1.3 Subgroup 3. For this subgroup, the first and each hundredth production unit thereafter shall be selected.

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TABLE V - GROUP C INSPECTION

INSPECTION	REQ PARA	INSP PARA
Subgroup I		
Low Temperature	3.6.1.1	4.9 1.1
High Temperature	3.6.1.2	4.9.1.2
Humidity	3.6 2	4.9.2
Altitude	3.6.3	4.9.3
Vibration	3.6.4.1	4.9.4.1
Shock, Hi Impact	3.6.4.2	4.9.4.2
Shock, Bench Handling	3.6.4.3	4.9.4.3
Vibration, Secure Cargo	3.6.4.4	4.9.4.4
Bounce, Vehicular	3.6.4.5	4.9.4.5
Bounce, Loose Cargo	3.6.4.6	4.9.4.6
Immersion	3.6.11	4.9.11
Shock Drop	3.6.13	4.9.13
EMI	3 7.1	4.10
TEMPEST	3.8	4.11
Subgroup 2		
Temperature Shock	3.6.7	4.9.7
Sunshine	3.6.8	4.9.8
Rain	3.6.9	4.9.9
Explosive Atmosphere	3.6.10	4.9.10
Salt Fog (See Note 1)	3.6.5	4.9.5
Fungus (See Note 1)	3.6.6	4.9.6
Dust (See Note 1)	3.6.12	4.9.12
Subgroup 3		
TSEC/KY-57 System Test	3.15	4.15

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent.

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4.5.3.2 Order of Inspection Within Group C. Group C inspection shall be performed on sample units in the order specified in 4.5.3.1.

4.5.3.3 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract.

4.5.4 Reinspection of Conforming Group B and C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

4.5.5 Group D Inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

TABLE VI - GROUP D INSPECTION

INSPECTION	REQ PARA	INSPE PARA
Reliability	3.11	4.14

4.5.5.1 Sampling for Group D Inspection. For reliability purposes a lot shall be defined as one month's production.

4.5.5.2 Group D Failures. Actions required relative to Group D failures shall be as specified in the contract.

4.5.5.3 Reinspection of Conforming Group D Sample Units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

4.6 Visual and Mechanical Inspection. (See 3.10, 3.19) Equipment shall be examined for the defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with those portions of 3.10 (safety engineering) which can be determined visually.

4.7 Preconditioning. (See 3.4)

4.7.1 Bounce Preconditioning. (See 3.4.1) Each unit, in its non-operating condition, shall be placed in its normal operating position on the table of the package tester Type 1000 SC, as made by the L.A.B Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. Each unit of each lot shall be subjected to this precondition. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

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4.7.2 Electrical Preconditioning (Burn-In). (see 3.4.2) Each CMSC unit shall be subjected to the test conditions cited in 4.7.2.1 and shall meet the requirements of 3.4.2. Performance testing as specified in 4.8.4 shall be performed at the end of the first 24 hours and at the completion of the test.

4.7.2.1 Test Conditions. Testing shall be conducted under the conditions of Test Level C of MIL-STD-781B. The duty cycle shall be 50% transmit and 50% receive.

4.8 Electrical Tests.

4.8.1 Standard Conditions. Unless otherwise specified all tests shall be performed at 24 VDC $\pm$1 VDC from an external source. Tests shall be conducted at room ambient temperature, pressure and humidity per MIL-STD-810 unless otherwise specified.

4.8.2 Standard Test Equipment. Standard test equipment is indicated in test figures by manufacturer and model number. In all cases, equivalent substitutes may be used with prior approval of the Contracting Officer's Representative. The required test equipment is indicated in Table VII. Standard test equipment shall be provided by the contractor.

4.8.3 Special Test Equipment. The CMSC test set which is specified in Table VII and described in Figure 1, or its equivalent, shall be provided by the contractor.

4.8.4 Operational Tests.

4.8.4.1 Transmit 16 Kb digital data mode test. (See 3.5.1). This test will verify that the CMSC will provide the functions of digital data transmission including KY keying, data flag generator, data input clocking, double PTT interrupt, message resumption after interrupt, priority mode using LDDMC and message termination. Digital mode KY and Local Data Connector interfaces will also be verified during this test.

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TABLE VII - TEST EQUIPMENT REQUIREMENTS

	Test Equipment
1 ea	CMSC Test Set (including a CMSC)
1 ea	28 VDC Power Supply, 500 mA, Power Designs 3650-S
1 ea	Volt-Ohmmeter, Weston 663
1 ea	AC Voltmeter, HP 400 EL
1 ea	Audio Oscillator, HP 651B
1 ea	Oscilloscope, Tektronix Type 422
1 ea	Power Supply - 15 V at 100 mA, HP 721A
1 ea	Power Supply - 5 V at 30 mA, HP 721A

4.8.4.1.1 Test Using Local Digital Mode Switch Position. (See 3.5.1)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions.

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
TSCMSC Power ----------Off
AO/DDO ----------Off
TSCMSC AO/DDO ----------DDO
Data Polarity ----------Off
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------CMSC LDDI
Test Sig Source ----------16Kb Data
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------TSCMSC/LAO/DDO (1)[1]

1 The number parenthesis identifies the CMSC Test Set monitor switch position.

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c. Verify that the test set and test equipment power is on. Perform all remaining test operations in the sequence noted belowed.

d. Set the CMSC Mode switch to "D".

e. Set the TSCMSC Mode switch to "D" and the TSCMSC power switch to "on Battery".

f. Turn CMSC power switch to "on Battery" and record CMSC DC current. (Limit: 107 mA max)

g. Turn LPTT switch "on".

h. Turn TSCMSC Power Up switch "on".

i. Turn Clock switch "on".

j. Verify that 16 Kb data is present on the oscilloscope.

k. Turn the Monitor switch to ADMC/DDI (2). Measure and record the plus and minus peak pulse voltage on the oscilloscope. (Limit: +6±1, -6±1V).

l. Turn the Monitor switch to the Test Sig Source (3). Measure and record the plus and minus peak pulse voltage on the oscilloscope. (Limit: +6±1, -6±1V).

m. Turn the monitor switch to PTT (12) and verify that PTT is at ground potential.†

n. Turn the monitor switch to DDMC (10) and verify that DDMC is at ground potential.

o. Set the monitor switch to LDMIC/DDCO (4). On the oscilloscope, measure and record the clock output level maximum and minimum peak voltage. (Limit: +6±1,-6±1V).

p. To verify operation of double PTT, set the monitor switch to TSCMSC/ LAO/DDO (1) and observe output data. Turn the HPTT switch to on twice within two seconds and hold the HPTT switch on after the second switching effort. Verify that data transmissions cease.

q. Turn the TSCMSC power up switch off, CLOCK off, HPTT off, and TSCMSC "Power Up" on. Turn on clock switch and note that data transmissions resume as indicated by the digital data on the oscilloscope.

r. Turn the monitor switch to ADMC/DDI (2). Then turn the LPTT switch off and note that data transmissions stopped as indicated by no digital data on the oscilloscope.

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† Ground potential is 0± 1V.

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4.8.4.1.2 Test Using LDDMC. (See 3.5.1, 3.5.11)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
TSCMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Off
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------CMSC LDDI
Test Sig Source ----------16 Kb Data
FSK Level ----------High
3900 Hz Tone ----------Off
Faud Baud Rate ----------600
Monitor Switch ----------TSCMSC/LAO/DDO (1)
TSCMSC AO/DDO ----------DDO

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to 3900 to verify that LDDMC controls mode priority. Perform all remaining test operations in the sequence noted below.

e. Set the TSCMSC Mode Switch to "D" and the TSCMSC power switch to "on" Battery".

f. Turn CMSC power switch to "on Battery" and turn LDDMC switch on.

g. Turn LPTT switch on.

h. Turn TSCMSC power switch on.

i. Turn clock switch on.

j. Verify that the digital data mode was established by verifying the presence of 16Kb data on the oscilloscope.

k. Turn the monitor switch to ADMC/DDI (2) and then turn the LPTT Switch off and note that data transmission stopped as indicated by no digital data on the oscilloscope.

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4.8.4.2 Receive 16Kb Digital Data Mode Test. (See 3.5.2) This test will verify operation of the CMSC functions associated with necessary digital data including KY power up detection, data flag detection, KY mode control, signal routing to LDDO, clock routing to LDDCO, message reception, recognition of KY power down and proper interfacing of the KY and local data connector.

4.8.4.2.1 Test Using Local Digital Mode Switch Position. (See 3.5.2)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Normal
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
TSCMSC Power ----------Off
TSCMSC AO/DDO ----------Off
Test Sig Routing ----------TSCMSC/LDDI
Test Sig Source ----------16 Kb Data
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------LAO/DDO (6)

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to "D".

e. Set the TSCMSC Mode Switch to "D" and TSCMSC Power switch to "on Battery".

f. Turn CMSC Power switch to "on Battery" and record CMSC DC current. (Limit: 107 mA max).

g. Set TSCMSC/LADMC/DDI switch to DDI.

h. Turn CMSC AO/DDO switch to DDO.

i. Turn TSCMSC LPTT switch to "on".

j. Turn CMSC power up switch "on".

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k. Turn Clock switch "on".

l. Verify that 16 Kb data is on the oscilloscope and record level. (Limit: +6±1,-6±1V).

m. Set Monitor switch to AO/DDO (7). On oscilloscope measure and record CMSC digital input level. (Limit: +6±1,-6±1V).

n. Set Monitor switch to TSCMSC LDMIC/DDCO (11). Record the clock level. (Limit: +6±1,-6±1V).

o. Set Monitor switch to LDMIC/DDCO (4). On oscilloscope measure and record CMSC clock output level. (Limit: +6±1,-6±1V).

p. Turn off "CMSC power up" and verify that the output as viewed on the oscilloscope is disabled.

4.8.4.2.2 Test Using LDDMC. (See 3.5.2, 3.5.11)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII, or their equivalents.

b. Preset the test set switches in the positions:

HPTT ----------Off
TSCMSC Power ----------Off
LPTT ----------Off
TSCMSC AO/DDO ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Normal
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------TSCMSC/LDDI
Test Sig Source ----------16 Kb Data
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------LAO/DDO (6)

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to A1 to demonstrate priority of LDDMC.

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e. Set the TSCMSC Mode Switch to "D" and TSCMSC power switch to "on Battery".

f. Turn CMSC power switch to "on Battery".

g. Set TSCMSC/LADMC/DDI switch to DDI.

h. Turn CMSC AO/DDO switch to DDO.

i. Turn LDDMC switch to on.

j. Turn TSCMSC LPTT switch to "on".

k. Turn CMSC power up switch "on".

l. Turn clock switch "on".

m. Verify that 16 Kb data is on the oscilloscope and record level. (Limit: $+6\pm1,-6\pm1$V).

n. Turn off "CMSC Power Up" and verify that the output as viewed on the oscilloscope is disabled.

4.8.4.2.3 Data Inversion Test. (See 3.5.2)

a. Connect the test set up as shown in Figure 1 using the test instruments shown in Table VII, or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
TSCMSC Power ----------Off
LPTT ----------Off
TSCMSC AO/DDO ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Normal
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LADMC ----------Off
TSCMSC/LDDMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------TSCMSC/LDDI
Test Sig Source ----------16 Kb Data
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------LAO/DDO (6)

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c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode switch to D.

e. Set the TSCMSC Mode switch to D and TSCMSC power switch to "on Battery".

f. Turn CMSC power switch to "battery".

g. Set Data Polarity switch to inverted.

h. Set TSCMSC/LADMC/DDI switch to DDI.

i. Turn CMSC AO/DDO switch to DDO.

j. Turn TSCMSC LPTT switch to on.

k. Turn CMSC Power up switch "on".

l. Turn Clock switch "on".

m. Verify that 16 Kb data is on the oscilloscope. Record Level. (Limit: +6±1,-6±1V).

n. Turn off "CMSC Power Up" and verify that the output as viewed on the oscilloscope is disabled.

4.8.4.3 Transmit Al Data Test. (See 3.5.3) This test will verify that the CMSC will provide for Al data transmission including KY keying, generation and transmission of data flag on DDI, transmission of data on DMIC, and proper interfacing of the local data and KY connectors for the Al data mode.

4.8.4.3.1 Test Using Local Al Mode Switch Position. (See 3.5.3)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off	TSCMSC/LDDMC ----------Off
LPTT ----------Off	TSCMSC/LADMC/DDI ----------Off
LDDMC ----------Off	Clock on/off ----------Off
LADMC/DDI ----------Off	Test Sig Routing ----------CMSC/LDMIC
CMSC Power Up ----------Off	Test Sig Source ----------Local Voice/Al
CMSC Power ----------Off	FSK Level ----------High
AO/DDO ----------Off	3900 Hz Tone ----------Off
Data Polarity ----------Off	FSK Baud Rate ----------600
TSCMSC Power Up --------Off	Monitor Switch ----------Test Signal Source (3)
TSCMSC Power ----------Off	
TSCMSC/LPTT ----------Off	TSCMSC AO/DDO ----------DDO

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c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to A1.

e. Set the TSCMSC Mode Switch to A1 and the TSCMSC power switch to "on Battery".

f. Turn CMSC power switch to "on Battery" and record CMSC DC current. (Limit: 107 mA max).

g. As a test signal use the external oscillator with its output set for a 1 KHz sine wave signal level as viewed on the output voltmeter to five millivolt RMS.

h. Turn LPTT on.

i. Turn TSCMSC Power up switch on.

j. Turn clock switch on. Set TSCMSC AO/DDO to AO and turn clock "off".

k. Turn the monitor switch to PTT (12) and note that PTT is at ground potential.

l. Turn the monitor switch to ADMC (2) and note that ADMC is at ground potential.

m. Switch the monitor switch to TSCMSC/LAO/DDO (1) and verify that the 1 KHz tone was transmitted as indicated on the oscilloscope.

n. Switch the monitor switch to DMIC/DDCO (5) on the output meter and oscilloscope measure and record the A1 output level. (Limit: 0.1 to 1000 mV).

o. Turn off the LPTT and note that A1 data transmission ceases.

4.8.4.3.2 Test Using ADMC. (See 3.5.3., 3.5.11)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ---------Qff
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ---------Off
TSCMSC Power Up -------Off
TSCMSC Power ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------CMSC/LDMIC
Test Sig Source ----------Local Voice/A1
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
TSCMSC AO/DDO ----------DDO
Monitor Switch ----------Test Signal Source (3)

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c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to D.

e. Set the TSCMSC Mode Switch to A1 and the TSCMSC power switch to "on Battery".

f. As a test signal, use the external oscillator with its output set for a 1 KHz sine wave. Adjust the sine wave signal level as viewed on the output voltmeter and oscilloscope to five millivolts RMS. Turn CMSC power to "on Battery".

g. Turn the LADMC switch on.

h. Turn LPTT on.

i. Turn TSCMSC power up switch on.

j. Turn clock switch on.

k. Turn the monitor switch to PTT (12) and note that PTT is at ground potential.

l. Turn the monitor switch to ADMC (2) and note that ADMC is at ground potential.

m. Turn off clock. Set TSCMSC AO/DDO to AO. Switch the monitor switch to TSCMSC/LAO/DDO (1). Verify that the 1 KHz tone was transmitted as indicated on the oscilloscope.

n. Turn off the LPTT and note that A1 data transmission ceases.

4.8.4.4 Receive A1 Data Mode Test . (See 3.5.4) This test will verify operation of CMSC A1 data reception including KY power up detection, KY mode control, data flag detection, message routing, KY power down and proper interfacing with the local data and KY connectors.

4.8.4.4.1 Test Using Local A1 Mode Switch Position. (See 3.5.4)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

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HPTT ----------Off
TSCMSC Power ----------Off
LPTT ----------Off
TSCMSC AO/DDO ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Normal
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------TSCMSC/LDMIC
Test Sig Source ----------Wireline Voice (for Al)
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------Test Sig Source (3)

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to Al.

e. Set the TSCMSC Mode Switch to Al and the TSCMSC power switch to "on battery". Set the external oscillator frequency to a 1000 Hz sinewave. Using the output meter and oscilloscope, adjust the signal level to 1.5 volt RMS. Change monitor switch to LAO/DDO (6) position.

f. Turn CMSC power switch to "on Battery" and record CMSC DC current.

g. Turn CMSC AO/DDO switch to DDO to receive data flag.

h. Turn TSCMSC LPTT switch to "on".

i. Turn CMSC Power Up switch "on".

j. Turn Clock switch "on".

k. After 3 seconds turn clock off and switch CMSC AO/DDO switch to AO.

l. Verify that Al data is on the oscilloscope. Record level. (Limit: 3V RMS +3 dB)
-6

m. Turn off TSCMSC/LPTT and verify that the output, as viewed on the oscilloscope, is disabled.

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4.8.4.4.2 Test Using LADMC Priority. (See 3.5.4, 3.5.11)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Normal
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC Power ----------Off
TSCMSC AO/DDO ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------TSCMSC/LDMIC
Test Sig Source ----------Wireline Voice (for A1)
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------LAO/DDO

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode to A2.

e. Set the TSCMSC Mode Switch to A1 and the TSCMSC power to "on Battery".

f. Turn CMSC power switch to "on Battery".

g. Turn CMSC AO/DDO switch to DDO.

h. Turn TSCMSC LPTT switch to "on".

i. Turn LADMC on. Turn CMSC Power Up switch "on".

j. Turn clock switch "on".

k. After 3 seconds, turn clock off and switch CMSC AO/DDO switch to AO.

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l. Verify that data is on the oscilloscope. Record level. (Limit: 3V RMS +3 dB)
-6

m. Turn off TSCMSC/LPTT and verify that the output as viewed on the oscilloscope is disabled.

4.8.4.5 Transmit A2 Data Mode Test. (See 3.5.5) This test will verify operations of the CMSC functions of voice/data detection, data flag generation, KY Keying, message transmission, VOX Keying, 3900 Hz keying, wireline A2 interfaces and local data connector A2 interfaces.

4.8.4.5.1 Test Using Local A2 Mode Switch Position. (See 3.5.5)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Off
TSCMSC Power Up ----------Off
TSCMSC Power ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------CMSC LDMIC
Test Sig Source ----------Local A2
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------TSCMSC/LAO/DDO (1)
TSCMSC AO/DDO ----------DDO

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to A2.

e. Set the TSCMSC Mode Switch to A2 and the TSCMSC power switch to "on Battery".

f. Turn CMSC power switch to "on Battery" and record the CMSC DC current. (Limit: 107 mA max).

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g. Turn LPTT "on".

h. Turn TSCMSC power switch "on".

i. Turn clock switch "on" for about 3 seconds then turn "off".

j. Verify that TACFIRE data is present on the oscilloscope.

k. Turn the monitor switch to PTT (12) and note that PTT is at ground potential.

l. Turn the monitor switch to DDMC (10) and note that DDMC is at ground potential.

m. Set monitor switch to TSCMSC LAO/DDO (1). Change FSK Baud data from 600 to 1200. Turn LPTT, TSCMSC power up off, then turn them on in the same order. Turn clock "on". Verify that on the oscilloscope the TACFIRE data is now being transmitted at 1200 baud. Then set monitors to PTT (12).

n. Turn LPTT off and verify that data transmission stops as indicated by a PTT (12) signal which is +5 VDC ±1 VDC.

4.8.4.5.2 Test Using 3900 Hz Keying Tone. (See 3.5.5)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Off
TSCMSC Power Up ----------Off
TSCMSC Power ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------CMSC Wireline
Test Sig Source ----------Remote A2
FSK Level ----------Low
3900 Hz Tone ----------Off
Monitor Switch ----------TSCMSC/LAO/DDO (1)
TSCMSC AO/DDO ----------DDO

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c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to 3900.

e. Set the TSCMSC Mode Switch to A2 and the TSCMSC power switch to "on Battery".

f. Turn CMSC power switch to "on Battery" and record the CMSC DC current. (Limit: 107 mA max).

g. Turn the 3900 Hz keying tone on.

h. Turn TSCMSC power up switch on.

i. Turn clock switch on for about 3 seconds, then "off".

j. Verify that TACFIRE data is present on the oscilloscope.

k. Turn the monitor switch to PTT (12) and note that PTT is at ground potential.

l. Turn the monitor switch to DDMC (10) and note that DDMC is at ground potential.

m. Set monitor switch to TSCMSC LAO/DDO (1). Switch the FSK signal level switch from low to high and note that data transmission continue as viewed on the oscilloscope.

n. Change FSK Baud Rate from 600 to 1200. Turn 3900 tone off and TSCMSC power up off. Then turn these switches on in the same order, then turn clock on. Verify that on the oscilloscope the TACFIRE data is now being transmitted at 1200 baud.

o. Turn 3900 tone switch off and verify that data transmission stops as indicated by absence of data on the oscilloscope.

4.8.4.5.3 Test Using VOX Keying. (See 3.5.5)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

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HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Off
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC Power ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------CMSC Wireline
Test Sig Source ----------Remote A2
FSK Level ----------Low
3900 Baud Rate ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------TSCMSC/LAO/DDO (1)
TSCMSC AO/DDO ----------DDO

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to VOX.

e. Set the TSCMSC Mode Switch to A2 and the TSCMSC power switch to "on Battery".

f. Turn CMSC power switch to "on battery" and record the DC current. (Limit: 107 mA max).

g. Turn TSCMSC Power up switch on. The keying signal will be established by the built in VOX detector.

h. Turn clock switch on for about 3 seconds and then "off".

i. Verify that TACFIRE data is present on the oscilloscope.

j. Turn the monitor switch to PTT (12) and note that PTT is ground potential.

k. Turn the monitor switch to DDMC (10) and note that DDMC is at ground potential.

l. Set monitor switch to TSCMSC LAO/DDO (1). Switch the FSK signal level switch from low to high and note that data transmissions as viewed on the oscilloscope continue.

m. Change FSK Baud Rate from 600 to 1200. Turn off TSCMSC power up and change the position of the routing switch to LDMIC. Turn TSCMSC power up on, the routing switch to CMSC wireline and then turn the clock switch to on. Verify that on the oscilloscope, FSK Data is being transmitted at 1200 baud.

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n. Turn routing switch to LDMIC and verify that data transmission stops as indicated by absence of data on the oscilloscope.

4.8.4.6 Receive A2 Data Mode Test. (See 3.5.6) This will verify operations of CMSC A2 data reception including the functions of KY power up detection, data flag detection, message routing, message reception, no LDDCO output, recognition of KY power down and proper interfacing of the A2 data to the wireline, local data and KY connector.

4.8.4.6.1 Test Using Local A2 Mode Switch Position. (See 3.5.6)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
TSCMSC Power ----------Off
TSCMSC AO/DDO ----------DDO
AO/DDO ----------Off
Data Polarity ----------Normal
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------TSCMSC/LDMIC
Test Sig Source ----------Local A2
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------LAO/DDO (6)

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode switch to A2.

e. Set the TSCMSC Mode switch to A2 and the TSCMSC power to "on Battery".

f. Turn CMSC power switch to "Battery" and record CMSC DC current. (Limit: 107 mA max).

g. Turn CMSC AO/DDO switch to DDO.

h. Turn TSCMSC LPTT switch to "on".

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i. Turn CMSC power up switch "on".

j. Turn clock switch "on".

k. Verify that A2 data is on the oscilloscope. Record Level (Limit: .21V±.07V).

l. Set monitor switch to AO/DDO (7). On oscilloscope measure and record CMSC digital input level. (Limit: +6±1,-6±1V).

m. Set monitor switch toLDMIC/DDO (4). On oscilloscope verify that there is no clock output.

n. Set monitor switch to AO/DDO (7). Turn off clock TSCMSC/LPTT and CMSC power up and verify that the output as viewed on the oscilloscope is disabled.

4.8.4.6.2 Test Using 3900 Remote Mode Switch Position. (See 3.5.6)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Normal
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------TSCMSC/LDMIC
Test Sig Source ----------Local A2
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------CMSC Wireline (8)
TSCMSC Power ----------Off
TSCMSC/AO/DDO ----------Off

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to 3900.

e. Set the TSCMSC Mode switch to A2 and the TSCMSC power switch to "on Battery".

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f. Turn CMSC Power switch to "on Battery".

g. Turn CMSC AO/DDO switch to DDO.

h. Turn TSCMSC LPTT switch to "on".

i. Turn CMSC Power Up switch "on".

j. Turn Clock switch "on" for approximately 3 seconds then turn "off".

k. Verify that A2 data is on the oscilloscope. Record Level. (Limit: OdBm ±4dB).

l. Set Monitor switch to AO/DDO (7). On oscilloscope measure and record CMSC digital input level. (Limit: +6±1,-6±1V).

m. Set Monitor switch to LAO/DDO (6). On oscilloscope verify that there is no digital output.

n. Set Monitor switch to wireline (8). Turn off TSCMSC/LPTT and CMSC power up and verify that the output as viewed on the oscilloscope is disabled.

4.8.4.6.3 Test Using Remote VOX Mode Position. (See 3.5.6)

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
TSCMSC Power ----------Off
TSCMSC AO/DDO ----------Off
AO/DDO ----------Off
Data Polarity ----------Normal
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------TSCMSC/LDMIC
Test Sig Source ----------Local A2
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------CMSC Wireline (8)

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c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to VOX.

e. Set the TSCMSC Mode Switch to A2 and the TSCMSC power switch to "on Battery".

f. Turn CMSC power switch to "on Battery". (Limit: 107 mA max).

g. Turn CMSC AO/DDO switch to DDO.

h. Turn TSCMSC LPTT switch to "on".

i. Turn CMSC Power Up switch "on".

j. Turn Clock switch "on" for about 3 seconds and then turn "off".

k. Verify that A2 data is on the oscilloscope. Record level. (Limit: 0 dBm ±4dB).

l. Turn off TSCMSC/LPTT and CMSC Power Up and verify that the output as viewed on the oscilloscope is disabled.

4.8.4.7 Transmit Voice. (See 3.5.7)

4.8.4.7.1 Transmit Wireline Voice Test. (See 3.5.7.1) This test will verify operations of voice transmission including proper voice/data detection, no data flag generation, interfacing with wireline and KY connector, frequency response, VOX keying 3900 Hz keying and message termination. The following test first uses 3900 Hz keying.

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off	TSCMSC/LADMC/DDI ----------Off
LPTT ----------Off	Clock on/off ----------Off
LDDMC ----------Off	Test Sig Routing ----------CMSC Wireline
LADMC/DDI ----------Off	
CMSC Power Up ----------Off	TSCMSC Power ----------Off
CMSC Power ----------Off	TSCMSC AO/DDO ----------Off
AO/DDO ----------Off	Test Sig Source ----------Wireline Voice
Data Polarity ----------Off	FSK Level ----------High
TSCMSC Power Up --------Off	3900 Hz Tone ----------Off
TSCMSC/LPTT ----------Off	FSK Baud Rate ----------600
TSCMSC/LDDMC ----------Off	Monitor Switch ----------Test Signal (3)

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c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to 3900.

e. Turn the CMSC power switch to "on battery" and record the CMSC DC current. (Limit: 107 mA max).

f. As a test signal, use the external oscillator with its output set for a 1 kHz since wave. Adjust the sine wave signal level as viewed on the output voltmeter/oscilloscope to 0.79 volts RMS.

g. Turn the 3900 Hz tone on.

h. Turn the monitor switch to PTT (12) and note that PTT is at ground potential.

i. Set the monitor switch to CMSC DMIC/DDCO (5) and verify that the 1 kHz tone is being transmitted as indicated on the oscilloscope. Measure and record the 1 kHz tone level. (Limit: 0.1 to 1000 mV).

j. Turn off the 3900 Hz tone. Verify that the 1000 Hz tone transmissions cease as observed on DMIC.

4.8.4.7.2 Test Using VOX. (See 3.5.7.1)

a. Connect the test setup as shown in Figrue 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

TSCMSC AO/DDO ----------Off
TSCMSC Power ----------Off
HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------CMSC Wireline
Test Sig Source ----------Wireline Voice
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------Test Signal (3) Source

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c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to VOX. Turn the CMSC power switch to "on" battery.

e. As a test signal, use the external oscillator with its output set for a 1 kHz sine wave. Adjust the sine wave signal level as viewed on the output voltmeter/oscilloscope to 0.79 volts RMS.

f. Turn the monitor switch to PTT (12) and note that PTT is at ground potential.

g. Set the monitor switch to CMSC DMIC/DDCO (5) and verify that the 1 kHz tone is being transmitted as indicated on the oscilloscope. Measure and record the 1 kHz tone level. (Limit: 0.1 to 1000 mV).

4.8.4.7.3 Transmit Local Voice Test. (See 3.5.7.2) This test will verify operation of voice transmission via the local data connector LDMIC including proper voice/data detection, no data flag generation, interfacing with local data and KY connectors, LPTT keying and message termination.

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Off
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off
Test Sig Routing ----------CMSC LDMIC
Test Sig Source ----------Local Voice/Al
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------Test Signal Source (3)
TSCMSC Power ----------Off
TSCMSC AO/DDO ----------Off

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c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to A2.

e. Turn the CMSC power switch to "on battery" and record the CMSC DC current. (Limit: 107 mA max).

f. As a test signal, use the external oscillator with its output set for a 1 kHz sine wave. Adjust the sine wave signal level as viewed on the output voltmeter/oscilloscope to five millivolts RMS.

g. Turn the LPTT switch on.

h. Turn the monitor switch to PTT (12) and note that PTT is at ground potential.

i. Set the monitor switch to CMSC DMIC/DDCO (5) and verify that the 1 kHz tone is being transmitted as indicated on the oscilloscope. Measure and record the 1 kHz tone level. (Limit: 0.1 to 1000 mV).

j. Turn off the LPTT switch. Verify that the 1000 Hz tone transmissions cease as observed on DMIC.

4.8.4.7.4 Transmit Handset Voice Mode (Power On) (See 3.5.7.3) This test will verify operation of voice transmission via the handset with CMSC power on. Functions tested will include KY keying, no data flag generation and interfacing with the KY and handset connector.

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

TSCMSC AO/DDO ----------Off
TSCMSC Power ----------Off
HPTT ----------Off
LPTT ----------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up ----------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity ----------Off
TSCMSC Power Up ----------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ----------Off
TSCMSC/LADMC/DDI ----------Off
Clock on/off ----------Off

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Test Sig Routing ----------HXMT
Test Sig Source ----------Handset Voice
FSK Level ----------High
3900 Hz Tone ----------Off
FSK Baud Rate ----------600
Monitor Switch ----------Test Signal Source (3)

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to D.

e. Turn the CMSC power to "on battery" and record the CMSC DC current. (Limit: 107 mA max).

f. As a test signal, use the external oscillator with its output set for a 1 kHz sine wave. Adjust the sine wave signal level as viewed on the output voltmeter/oscilloscope to five millivolts RMS.

g. Turn the HPTT switch on.

h. Turn the monitor switch to PTT (12) and note that PTT is at ground potential.

i. Set the monitor switch to CMSC DMIC/DDCO (5) and verify that the 1 KHz tone is being transmitted as indicated on the oscilloscope. Measure and record the 1 kHz tone level. (Limit: 0.1 to 1000 mV).

j. Turn off the HPTT switch. Verify that the 1000 Hz tone transmissions cease as observed on DMIC.

4.8.4.8 Receive Voice Mode Test. (See 3.5.8) This test will verify operation of CMSC voice reception including power up detection, proper KY mode control after no data flag detection, signal routing, recognizing power down and proper interfacing with the KY, wireline, and local data connectors.

a. Connect the test setup as shown in Figure 1 using the test instruments shown in Table VII or their equivalents.

b. Preset the test set switches in the following positions:

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HPTT ----------Off
TSCMSC Power ---------Off
LPTT ----------Off
TSCMSC AO/DDO --------Off
LDDMC ----------Off
LADMC/DDI ----------Off
CMSC Power Up --------Off
CMSC Power ----------Off
AO/DDO ----------Off
Data Polarity --------Normal
TSCMSC Power Up ------Off
TSCMSC/LPTT ----------Off
TSCMSC/LDDMC ---------Off
TSCMSC/LADMC/DDI -----Off
Clock on/off ---------Off
Test Sig Routing -----TSCMSC/LDMIC
Test Sig Source ------Wireline Voice (for A1)
FSK Level ----------High
3900 Hz Tone ---------Off
FSK Baud Rate --------600
Monitor Switch -------CMSC AO/DDO (7)

c. Verify that the test set and test equipment power is on.

d. Set the CMSC Mode Switch to A2.

e. Set the TSCMSC Mode Switch to A2 and the TSCMSC power switch to "on Battery". Use the 1000 Hz tone from the external oscillator.

f. Turn CMSC power switch to "on Battery" and record CMSC current. (Limit: 107 mA max).

g. Turn CMSC AO/DDO switch to AO.

h. Turn TSCMSC LPTT switch to "on".

i. Turn CMSC Power Up Switch "on".

j. Adjust the tone level to 1.5 volts RMS as seen on the output meter/ oscilloscope. Turn clock on.

k. Set monitor switch to HREC (9). Verify presence of 1000 Hz tone as seen on oscilloscope. Carefully record level. (Limit: 1.0V +3 dB)
-6

l. Set monitor switch to LAO/DDO (6). Turn off clock. Verify presence of 1000 Hz tone as seen on oscilloscope. Record level. (Limit: 3V +3 dB)
-6

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m. Turn "CMSC Power Up" off, set CMSC mode switch to 3900, then turn "CMSC Power Up" on. Turn on the clock for about 3 seconds then clock off. Set monitor switch to Wireline (8) verify presence of 1000 Hz tones as seen on oscilloscope. Record Level. (Limit: o dBm ±4 dB).

n. Set monitor switch to AO/DDO (7). Change tone frequency to 300 Hz. Adjust for 1.5 volts RMS as seen on output meter/oscilloscope. Set monitor switch to wireline (8). Record level. (Limit: 0 dBm ±4 dB).

o. Repeat step m for a tone frequency of 2700 Hz. Record level. (Limit: 0 dBm ±4 dB).

p. Turn off CMSC Power Up. Turn off Tone Oscillator. Disconnect wireline cable from CMSC. Measure DC resistance across wireline terminals. (Limit: 100K min.).

q. Turn off LPTT.

4.8.4.9 Plain Text Reception. (See 3.5.9)

a. Repeat test procedure 4.8.4.8, steps a, b, c, d, e, f, g, and h.

b. Set monitor switch to LAO/DDO (6) and verify presence of 1000 Hz test tone.

c. Set CMSC mode switch to VOX and monitor switch to wireline (8) and verify presence of 1000 Hz test tone.

4.8.4.10 Power off, Bypass Mode Test. (See 3.5.10) With no signal or power cable connections to the CMSC, measure continuity between the following connections.

a. KY Pin A to KY connector Pin D (<150 ohms)

b. Audio Pin B to KY connector Pin B (<150 ohms)

c. Audio Pin C to KY connector Pin C (<10 ohms)

d. Audio Pin A to Audio connector Pin D (<5K ohms)

4.9 Service Condition Tests. (See 3.6)

4.9.1 Temperature Test. (See 3.6.1)

4.9.1.1 Low Temperature. (See 3.6.1.1) The equipment shall be tested in accordance with Test Method 502.1, Procedure I of MIL-STD-810. Storage temperature shall be minus 70 degrees F. (Minus 57 degrees C). Storage Time shall be 24 hours minimum. Lowest operating temperature shall be minus 60 degrees F (minus 51 degrees C). After stabilization at the operating temperature, the equipment shall operate without degradation in specified performance within 5 minutes after power application. Performance tests, as specified in 4.8.4, shall be made during testing, as specified in Test Method 502.1. After testing, performance tests, as specified in 4.8.4, shall be made.

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4.9.1.2 High Temperature. (See 3.6.1.2) The equipment shall be tested in accordance with Test Method 501.1, Procedure II of MIL-STD-810. Highest Operating temperature shall be plus 160 degrees F (plus 71 degrees C). Performance tests, as specified in 4.8.4 shall be made during testing, as specified in Test Method 501.1. After testing, performance tests, as specified in 4.8.4 shall be made.

4.9.2 Humidity Test. (See 3.6.2) The equipment shall be tested in accordance with Test Method 507.1, Procedure II of MIL-STD-810. Performance testing as specified in 4.8.4 shall be made during Step 4 and the last 5 hour period of each 48 hour cycle. At the conclusion of testing, performance testing as specified in 4.8.4 shall be made.

4.9.3 Low Pressure (Altitude) Test. (See 3.6.3) The equipment shall be tested in accordance with Test Method 500.1, Procedure I of MIL-STD-810. In Step 2, the chamber altitude shall be 15,000 feet above sea level. Performance tests, as specified in 4.8.4.1.1, shall be made during testing, as specified in Test Method 500.1. After testing, performance tests, as specified in 4.8.4 shall be made.

4.9.4 Vibration, Shock and Bounce Tests. (See 3.6.4)

4.9.4.1 Vibration Test. (See 3.6.4.1) The equipment shall be tested in accordance with Test Method 514.2, Procedure VIII of MIL-STD-810. Curve W of Figure 514.2-6 shall be used. Performance testing as specified in 4.8.4 shall be made after testing as specified in Test Method 514.2.

4.9.4.2 Shock Test (Ballistic/Hi Impact). (See 3.6.4.2) The test shall be conducted on the "Shock Testing Machine for Lightweight Equipment" shown in Figure 1 of MIL-S-901. The CMSC with its vehicular Power Adapter to be tested shall be mounted on a Government furnished Mounting Base MT-4626()/URC. The equipment with its Mounting Base shall be secured in its normal operating position to the steel test plate by means of the same type fasteners used for vehicular installation of the equipment. The test shall consist of a total of nine blows; one each 1-foot blow, 3-foot blow, and a 5-foot blow on the back, side, and top of the test plate. As an alternate to re-orienting the plate for the blows on the side of the plate, an equivalent rotation of the equipment under test is permissible. Performance testing as specified in 4.8.4.1.1 shall be made during testing and paragraph 4.8.4 shall be made after testing.

4.9.4.3 Shock (Bench Handling) Test. (See 3.6.4.3) The equipment shall be tested in accordance with Test Method 516.2, Procedure V of MIL-STD-810. Performance testing as specified in 4.8.4 shall be made after testing as specified in Test Method 516.2.

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4.9.4.4 Vibration, Secure Cargo. (See 3.6.4.4) The equipment shall be tested per Method 514.2, Procedure X, Curve AX of MIL-STD-810. Performance testing as specified in 4.8.4 shall be made after testing per Method 514.2.

4.9.4.5 Bounce, Vehicular. (See 3.6.4.5) The equipment shall be tested per Method 514.2, Procedure IX, Part 2 of MIL-STD-810. Performance testing as specified in 4.8.4 shall be made after testing per Method 514.2.

4.9.4.6 Bounce, Loose Cargo Test/ (See 3.6.4.6) The equipment shall be tested in accordance with Test Method 514.2, Procedure XI, Part 2 of MIL-STD-810. Performance testing, as specified in 4.8.4, shall be made after testing.

4.9.5 Salt Fog Test. (See 3.6.5) The equipment shall be tested in accordance with Test Method 509.1 of MIL-STD-810. At the conclusion of the test the equipment shall be examined for evidence of visible corrosion. Performance testing as specified in 4.8.4 shall be made after testing as specified in Test Method 509.1. Failure of the equipment to meet the requirements of 3.6.5 shall constitute failure of this test.

4.9.6 Fungus Test. (See 3.6.6) The equipment, with all covers installed, shall be tested in accordance with Test Method 508.1, Procedure I of MIL-STD-810. There shall be abundant growth colonization (see 6.4.3) on 50 percent or more of the area control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the start of fungus test. Failure of the equipment to meet the requirements of 3.6.6 shall constitute failure of this test.

4.9.7 Temperature Shock. (See 3.6.7) The equipment shall be tested in accordance with Test Method 503.1, Procedure I of MIL-STD-810. Performance testing as specified in 4.8.4 shall be made after testing as specified in Test Method 503.1.

4.9.8 Sunshine. (See 3.6.8) The equipment shall be tested in accordance with Test Method 505.1, Procedure I of MIL-STD-810. The maximum CMSC case temperature shall not exceed +71 degrees C. Performance testing as specified in 4.8.4 shall be made after testing as specified in Test Method 505.1.

4.9.9 Rain. (See 3.6.9) The equipment shall be tested in accordance with Test Method 506.1, Procedure I of MIL-STD-810. After testing as specified in Test Method 506.1, the performance testing as specified in 4.8.4 shall be made.

4.9.10 Explosive Atmosphere. (See 3.6.10) The equipment shall be tested per Method 511.1, Procedure I of MIL-STD-810. During testing, no explosions are permitted.

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4.9.11 Immersion. (See 3.6.11) The equipment shall be tested per Method 512.1, Procedure I of MIL-STD-810.

4.9.12 Dust. (See 3.6.12) The equipment shall be tested per Method 510.1, Procedure I of MIL-STD-810. After testing as specified in Method 510.1, the performance testing as specified in 4.8.4 shall be made.

4.9.13 Shock Drop. (See 3.6.13) The equipment shall be tested per Method 516.2, Procedure II of MIL-STD-810. Performance testing as specified in 4.8.4 shall be made after testing per Method 516.2.

4.10 Electromagnetic Interference Test. (See 3.7) The equipment shall be tested for compliance to the requirements of 3.7. Test setups and test procedures shall conform to the measurement techniques of MIL-STD-462, Notice 3. Signal and data input and output terminals shall be terminated with simulated inputs and loads. Simulated loads shall have characteristic impedances similar to actual loads. All electromagnetic compatibility tests shall be performed with the vehicular power supply installed.

4.10.1 Bonds, Grounds and Connectors. (See 3.7.2 and 3.7.3) Prior to the performance of EMI tests, the bonding impedance requirements of 3.7.2 and 3.7.3 shall be verifed. Bonding impedance of the following items shall be measured with a Shallcross Meter Model 670A or equal:

a. All connector shells to chassic frame.

b. All surfaces requiring EMI gaskets.

c. All switches, meters, hinge fasteners, transformers, iron core inductors, and all mechanical mating surfaces which become exposed during maintenance and repair.

4.11 Tempest. (See 3.8) The equipment shall be tested as specified in a contractor prepared, Government approved test plan.

4.12 Inspection for Dimensional Interchangeability. (See 3.9 and 3.13) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.9 and 3.13. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

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a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, locking pins, fasteners, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.13 Electrical Subassembly Test. (See 3.12) Plug-in subassemblies, including spares, shall meet the requirements of their respective test requirements specification drawings.

4.14 Reliability. (See 3.11)

4.14.1 First Article Reliability Acceptance Test. A random sample of 12 equipments shall be subjected to reliability testing in accordance with the accept/reject criteria of Test Plan XXII of MIL-STD-781B.

4.14.2 Production Reliability Qualification Test. A random sample of 12 equipments shall be selected from the first month's production and subjected to reliability testing in accordance with the accept/reject criteria of Test Plan IVa of MIL-STD-781B.

4.14.3 Production Reliability Sampling Test. From each fourth lot or 1500 units, whichever is sooner, a random sample of 12 equipments shall be selected and subjected to reliability testing in accordance with accept/reject criteria of Test Plan IVa of MIL-STD-781B.

4.14.4 Reliability Test Conditions. All reliability testing shall be conducted under the conditions of Test Level C of MIL-STD-781B. The duty cycle shall be 50% transmit and 50% receive.

4.14.5 Performance Parameters to be Measured. During the reliability testing, performance testing as specified in 4.8.4 shall be performed on all units once a day. If a failure is detected, it shall be considered to have occurred at the time the equipment was last known to be operating properly.

4.14.6 Failure Definition. For reliability purpose a failure shall be defined as the inability of an equipment to perform its required function as specified in 3.5. This includes catastrophic as well as any event which causes departure from performance as required by this specification and the contractor's approved test procedure. Proper instrumentation shall be provided to insure the recognization of failure. Total test time per equipment shall be in accordance with 4.2.2.4 of MIL-STD-781B. Paragraph 5.4 through 5.10 inclusive of MIL-STD-781B shall also apply.

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4.14.7 Non-compliance. In case of Group D failure occurrence, procedures and requirements specified elsewhere in the contract shall be followed.

4.15 TSEC/KY-57 System Test. (See 3.15) The following confidence test shall be performed on the first and thereafter each hundredth production unit to verify that the CMSC unit performs its required interface and control functions with its parent equipment. A simulated communications link shall be configured with two (2) GFE TSEC/KY-57 speech security equipments operating back-to-back with a KY-57 back-to-back tester. A CMSC shall be attached to both the transmit and receive GFE TSEC/KY-57.

4.15.1 Transmission/Reception of 16 Kbit Digital Data. (See 3.15) A simulated synchronous 16Kbit data user shall initiate a keying signal to the CMSC data connector. As soon as digital data clock is made available to the simulated digital data user, an easil recognizable pattern shall be transmitted. The pattern shall be examined for completeness at the receive end.

4.15.2 Transmission/Reception of Local Analog 1 Data. (See 3.15) A simulated MIL-STD-188 data user shall initiate a keying signal followed by a delayed A1 data transmission. On the receive end, the analog data port of the CMSC shall be examined for analog data reception.

4.15.3 Transmission/Reception of Remote and Local Analog 2 Data. (See 3.15) The contractor shall test the CMSC in the local modes with simulated TACFIRE equipment or with GFE DMD units.

4.15.4 Transmission/Reception of Local/Remote Voice. (See 3.15) The contractor shall test the CMSC in all voice modes and verify the adequacy of end-to-end voice communications.

4.16 Weight. (See 3.14) A complete CMSC shall be weighed to determined compliance with 3.14.

4.17 Quality Conformance Inspection of Packaging. Preparation for a delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirement of Section 5.

4.18 Compliance With MIL-STD-188-114. (See 3.5.1.2) The digital data and receive clock outputs of the local data connector of the CMSC and equivalent outputs of the test set, Fig 1, shall be tested for compliance with paragraph 5.2 of MIL-STD-188-114.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B, or Commercial as specified (see 6.2b).

5.1.1 Level A.

5.1.1.1 Cleaning. Each CMSC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each CMSC shall be dried in accordance with the applicable procedures of MIL-P-116.

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5.1.1.3 Preservative Application. None required.

5.1.1.4 Unit Packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical Literature. Each technical literature shall be unit packed method 1C-1.

5.1.1.4.2 CMSC. Each CMSC shall be individually unit packed method III as follows: Cushion all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, Type CF, Class weather-resistant, variety SW, Grade W5c, designed to protect the unit from shock and vibration hazards during handling and transit. As an alternate, cushioning may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservation application and units packaging shall be as specified in 5.1.1.

5.1.3 Commercial Preservation. Preservation shall be in accordance with FED-STD-356.

5.2 Packing. Packing shall be Level A, B or Commercial as specified ((see 6.2b). Shipping containers for Level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of CMSCs, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PP-B-601, overseas type; PPP-B-621, style 4, class 2; or a PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitiate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

5.2.1.1 Utilization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

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5.2.2 Level B. A quantity of CMSCs, unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistance. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively, selection of the material and application specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film. When a single unit is shipped to a single destination, no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of CMSC's unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial Packing. Packing shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial Marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended Use. The CMSC shall be used where an interface control unit is required to control a TSEC/KY-57 with data supplied from a local or remote wire-line user and to route data to a proper interface point.

6.2 Ordering Data. Procurement documents should specify the following:

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a. Title, number and date of this specification and any amendment thereto.

b. Level A, Level B or commercial preservation and packing (see section 5).

c. When rough handling test is required.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required, test tools and running spares.

g. First article inspection. (see 3.1).

h. Length of cables (if any are required).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Definitions.

6.4.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main step or filament (hypha).

6.4.2 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4.3 Growth Colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundance growth.

6.4.4 Microbial Growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.4.5 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.4.6 Analog 1 (A1 Data). Any analog data which is contained within the pass band of 300 Hz to 2700 Hz.

6.4.7 Analog 2 (A2 Data). This is TACFIRE FSK data with mark/space of 1200/2400 Hz (±0.1%) at modulation rates of 600 or 1200 baud.

6.4.8 Digital Data. This represents data up to 16 Kb in accordance with paragraph 5.2 of MIL-STD-188-114.

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6.4.9 CMSC MIL-STD-188-114 Interface. This interface is in accordance with paragraph 5.2 of MIL-STD-188-114.

6.4.10 Wireline Information. Wireline information will consist of either voice or A2 data.

6.4.11 Local Information. This information consists of either voice, A1 data, A2 data or digital data applied to the local data connector.

6.4.12 Tubular Germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.5 Group C Inspection. (See paragraph 4.5.3.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the Contracting Officer on the adequacy of corrective action.

6.5.1 Initial Shipments. (See paragraph 4.5.3.3) Initial shipments of end items will be contingent only upon successful completion of all Group C tests.

6.6 Location of Operational Inspection. (See paragraph 4.8.4 and 4.5.1) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.7 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: C-10377W/GTC. The contractor shall apply for nomenclature in accordance with the applicable clause in the contract.

6.8 Verification Inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specifiction and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.9 Government Furnished Property List. The government shall furnish the below listed equipments in sufficient quantity to permit the contractor to perform required testing for the duration of the contract.

PART NO.	EQUIPMENT
ON241700	TSEC/KY-57
ON217339	TSEC/KY-57 Back to Back Tester with Cables
AN/PSG-2	Digital Message Device, TACFIRE
MT-4626	TSEC/KY-57 Equipment Mount
ON241845	Battery Case (Z-AlJ)
ON241780	Vehicular Power Adaptor HYP-57/TSEC
H-250	Handsets
ON190303	KYX-15/TSEC
ON241845	Battery Case (with Batteries)

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6.10 Conflicts. In case of conflict between a referenced specification and the production drawings defined in 2.0, the requirements of the drawings will take precedence since the design has been qualified to meet the drawing requirements. In case of conflict between this specification and any other document, this specification shall take precedence.

6.11 Interface Definitions

6.11.1 TSEC/KY-57 Interface. The CMSC interfaces to the TSEC/KY-57 only through the front panel audio connector J1, a 6-pin GC-283/U. The CMSC and TSEC/KY-57 interface characteristics are shown in Table VIII where inputs/outputs are with respect to the CMSC.

6.11.2 Local Data User Interface. The CMSC interfaces to local data users only through the front panel data connector, J3, a 6-pin GC-283/U. This interface is defined in Table IX. All transmit mode A1, A2, or voice signals are applied to Pin D. Voice is transmitted by the TSEC/KY-57 with the CMSC mode switch in the A2 data mode.

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TABLE VIII - FRONT PANEL TSEC/KY-57 CONNECTOR

SIGNAL NAME	CONNECTOR PIN TYPE	CONNECTOR PIN PIN	SIGNAL CHARACTERISTICS	NOMINAL IMPEDANCE SOURCE	NOMINAL IMPEDANCE LOAD
REFERENCE OR GROUND		A			
AQ/DDO	AUDIO IN	B	AUDIO OR ANALOG DATA IN 0.5 to 3.0V RMS, 300-2700 Hz	30 ohms	1K ohms
	DIGITAL IN		MIL-STD-188* LEVELS +1V CLOCK IN ON $\overline{\text{NEG}}$ EDGE	100 ohms	$\geq$6K ohms
PTT	PTT CONTROL	C	ACTIVE WHEN GROUNDED		
DMIC/ DDCO	AUDIO OUT	D	AUDIO OR ANALOG DATA OUT 10 mV +40 dB FOR WIRELINE $\overline{\text{VOICE}}$ 1 mV +20 dB FOR HANDSET, $\overline{\text{A1}}$ OR LOCAL VOICE 300 Hz TO 2.7 kHz	150 ohms	150 ohms
	CLOCK IN		MIL-STD-188* LEVELS $\pm$1V CLOCK INPUT	100 ohms	$\geq$6K ohms
DDMC	DIGITAL CONTROL OUT	E	GND FOR DIGITAL DATA MODE		100K ohms to -6.7V
ADMC/ DDI	ANALOG CONTROL OUT	F	GND FOR ANALOG DATA MODE		10K ohms to -6.7V
	DIGITAL DATA OUT		MIL-STD-188* LEVELS +1V DATA CLOCKS $\overline{\text{OUT}}$ ON POS. EDGE	100 ohms	$\geq$6K ohms

*MIL-STD-188-114, paragraph 5.2

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TABLE IX - LOCAL DATA CONNECTOR INTERFACES

SIGNAL NAME	CONNECTOR PIN TYPE	CONNECTOR PIN PIN	SIGNAL CHARACTERISTICS	NOMINAL IMPEDANCE SOURCE	NOMINAL IMPEDANCE LOAD
GROUND		A			
LAO/ DDO	AUDIO OUT	B	AUDIO OR ANALOG DATA OUT 300 Hz TO 2.7 kHz 0.14 TO 0.28V FOR A2 DATA 3V +6 dB FOR VOICE OR A1 DATA	30 ohms	1K ohms
	DIGI-TAL OUT		DIGITAL DATA +6V MIL-STD-188* LEVELS ±1V CLOCKED OUT ON POS. EDGE	100 ohms	≥6K ohms
LPTT	PTT	C	GND FOR ACTIVE PTT OPEN OTHERWISE	≤1K ohms	10K ohms to +5V
LDMIC/ DDCO	AUDIO IN	D	AUDIO OR ANALOG 1 DATA IN 1 mV ±12 dB 300 Hz TO 2.7 kHz ANALOG 2 DATA 5 mV	150 ohms	150 ohms
	CLOCK OUT		+6V MIL-STD-188 * LEVELS ±1V CLOCK	100 ohms	≥6K ohms
LDDMC	DIGITAL CONTROL INPUT	E	GND FOR DIGITAL DATA MODE		100K ohms to -6.7V
LADMC/ DDI	ANALOG CONTROL INPUT	F	GND FOR ANALOG DATA MODE		10K ohms to -6.7V
	DIGITAL DATA INPUT		+6V MIL-STD-188* LEVELS ±1V DATA CLOCKED IN ON NEG. EDGE	100 ohms	≥6K ohms

*MIL-STD-188-144, paragraph 5.2

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Table X CMSC's Audio Handset Connector Signal Interface Characteristics

Signal Name	Conn. Pin Type	Pin	Signal Characteristics	Nominal Impedance	
				Source	Load
GROUND		A			
HREC	AUDIO OUT	B	AUDIO SIGNAL OUTPUT 3 V ±6 dB 300 Hz to 2.7 kHz	150 ohms	1 K ohm
HPTT	PTT CONTROL	C	GND FOR ACTIVE PTT GND-OPEN-GND WITHIN 2 SECONDS FOR LOCAL VOICE INTERRUPT	≤1 K ohm	10 K ohms to +5 V
HXMIT	AUDIO	D	AUDIO SIGNAL INPUT FROM HANDSET 1 mW ±20 dB	150 ohms	150 ohms

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Table XI CMSC's Wireline Signal Interface Characteristics

INPUT	
Level	0 to -40 dBm
Z Load (ac)	600 ohms
Z Source (ac)	600 ohms
Frequency Range	300 Hz to 2.7 kHz
OUTPUT	
Level	0 ±4 dBm
Z Load (ac)	600 ohms
Z Source (ac)	600 ohms
Frequency Range	300 Hz to 2.7 kHz
KEYING SIGNALS	
3900 Hz Tone	(-15 dBm nominal)
	VOX (≥-40 dB)

Table XII CMSC's Power Connector Interface Characteristics

Signal Name	Conn. Pin Type		Signal Characteristics
GROUND		1	
INTLK	INTERLOCK	2	WIRED TO PIN 4
SPARE		3	
INTLK	INTERLOCK	4	WIRED TO PIN 2
+24 VDC	POWER INPUT	5	+22 TO +32 Vdc BATTERY OR VEHICULAR

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Figure 1. Test Diagram

and

Comment Sheet

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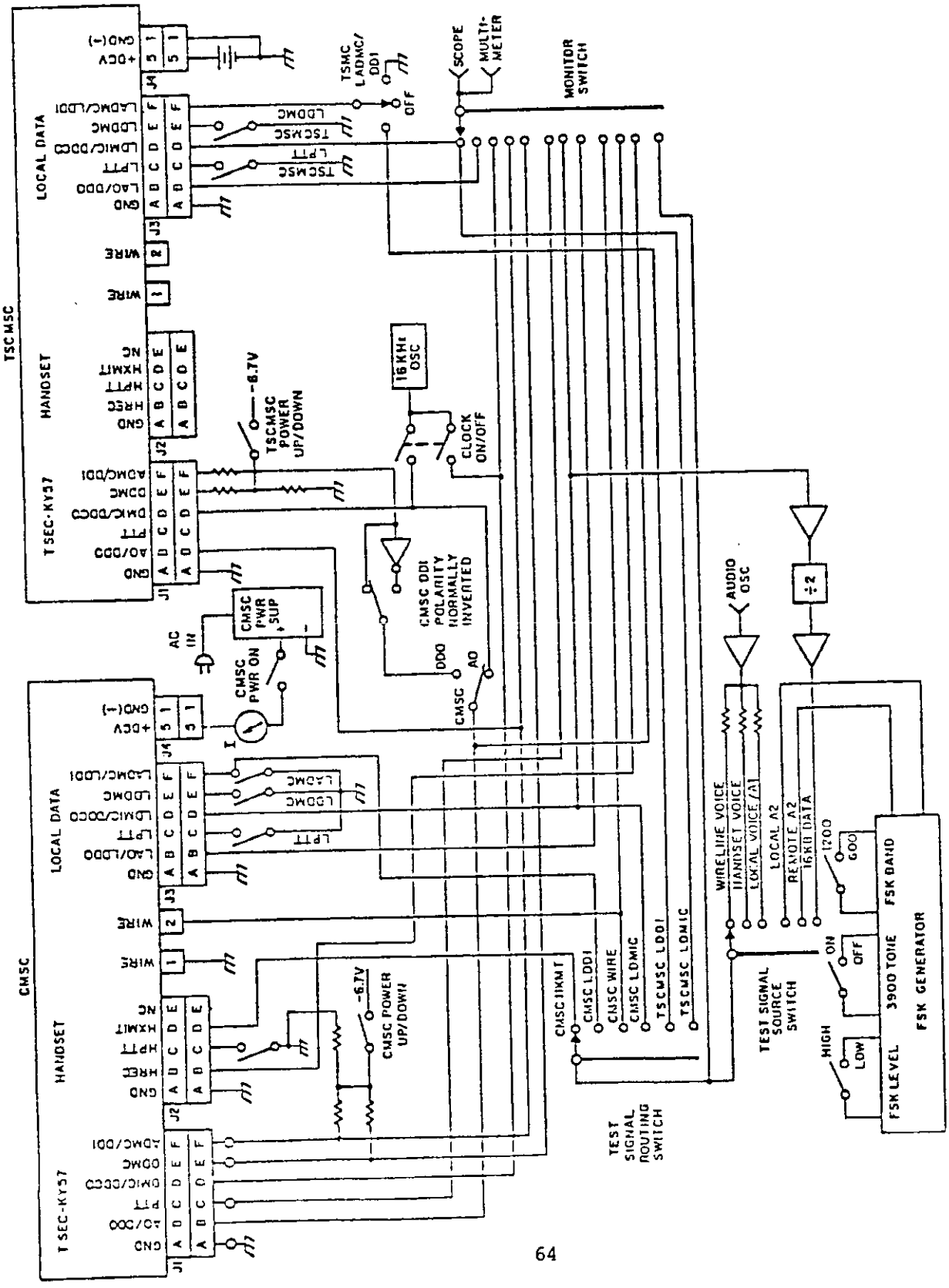


FIGURE 1. CMSC TEST DIAGRAM

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